

What is ...

OSU What Is...? Seminar

2025/07/15

Algebraic Complexity Theory?

Q: How do we differentiate between hard and easy problems

For what defn
of difficulty?

What kind?

A: Field of computational complexity

Algebraic Complexity Theory is concerned with the complexity of computation in the setting of polynomial rings.

Ex: $f(x) = \sum_{i=0}^d f_i \cdot x^i$, $f_i \in \mathbb{F}$. Written like this, computing $f_i \cdot x^i$ needs i multiplications $\Rightarrow d + \sum_{i=0}^d i = O(d^2)$ operations

Same as if f was factored

Consider Horner's Method [1819]

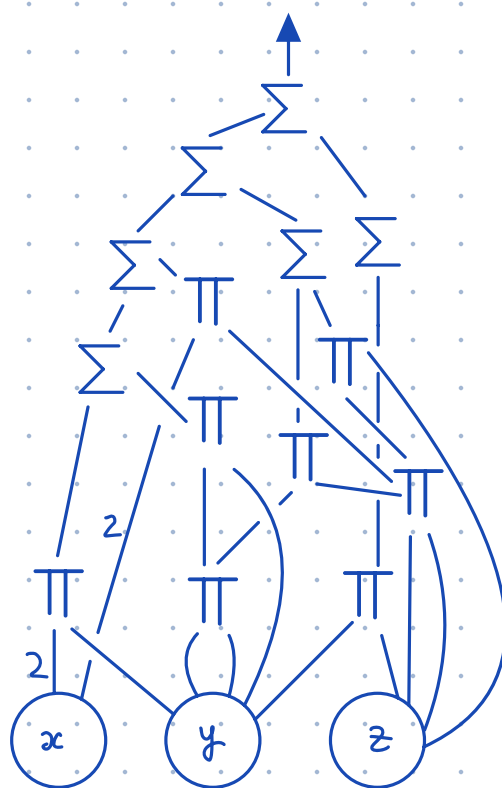
$f(x) = f_0 + x \cdot (f_1 + x \cdot (f_2 + \dots + x(f_{d-1} + x \cdot f_d) \dots))$ needs $O(d)$ ops

Thm [Ostrowski 1954]: Horner's Method is optimal!

Defn: An algebraic circuit over a polynomial ring $R[x_1, \dots, x_n]$ is a directed acyclic graph w/:

- Nodes of in-degree 0 labelled w/ variables x_i or scalars $\in R$
"inputs"
- Nodes labeled by Σ or Π
"sum and product gates"
- One node of out-degree 0
"output"
- Edges labelled by scalars $\in R$
"scalar multiplication"

Ex: $(2x + y^2 + z)(y + z^2) = 2xy + y^3 + yz + 2xz^2 + y^2z^2 + z^3$



size: 9

17

depth: 4

6

The size of a circuit is the # of vertices

The depth of a circuit is the length of the longest input $\rightsquigarrow$ output path

For $f \in \mathbb{R}[\bar{x}]$, $L(f) = \min \{s \mid \exists \text{ ckt of size } s \text{ computing } f\}$

Rmks:

1) No division, no equality/comparison checking

2) Common question types are upper/lower bounds on $L(f(x_1, \dots, x_n))$ under various restrictions on

- depth ($\Sigma \Pi$, $\Pi \Sigma$, $\Sigma \Pi \Sigma$, etc)

- in/out-degree

- homogeneity: if ckt C computes homog $f(\bar{x})$, do all gates compute homogeneous polynomials??

- Formulas: underlying graph is a tree

Lem: $xy + z \in \mathbb{C}[x, y, z]$ has no $\Pi\Sigma$ formula of any size

Pf: Spks towards contradiction that such a formula exists. Then this formula implies that $xy + z = \prod_{i=1}^k l_i$ w/ $l_i \in \mathbb{F}[x, y, z]$, $\deg(l_i) \leq 1$.

$\deg(xy + z) = 2 \Rightarrow k \geq 2$ and $\exists \deg 1 l_{i_1}, l_{i_2}$ dividing $xy + z$

But $xy + z$ is irreducible $\Rightarrow \Leftarrow$

Rmk:

1) $\Sigma\Pi$ formula exists (use monomial expansion)

2) With some work, can show no $\Pi\Sigma$ circuit exists

Q: Why care about such things?

A:

1) Let $G = (V, E)$ be a ^{simple} graph on $V = [n]$, n even. Does G have a perfect matching?

Let $T = (t_{ij})_{1 \leq i, j \leq n} = \begin{cases} x_{i,j} & \text{if } (i, j) \in E, i < j \\ -x_{i,j} & \text{if } (i, j) \in E, i > j \\ 0 & \text{if } (i, j) \notin E \end{cases}$

Tutte Matrix

Then G has a perfect matching $\Leftrightarrow \det(T) \neq 0$
 $\Leftrightarrow \text{perm}(T) \neq 0$

Small circuits for these polynomials $\Rightarrow$ fast algs for perfect matchings

2) [Agrawal - Kayal - Saxena 2002]: First deterministic primality test, developed using ideas from Polynomial Identity Testing, which is closely related to circuit complexity

3) Connections to P vs NP

The two stars of algebraic complexity are

$$\left. \begin{aligned} \det_n(X) &= \sum_{\sigma \in G_n} \text{sgn}(\sigma) \prod_{i=1}^n x_{i, \sigma(i)} \\ \text{perm}_n(X) &= \sum_{\sigma \in G_n} \prod_{i=1}^n x_{i, \sigma(i)} \end{aligned} \right\} \begin{array}{l} \text{Leibniz Formula} \\ O(n \cdot n!) \text{ circuits} \end{array}$$

What size circuits are possible for these?

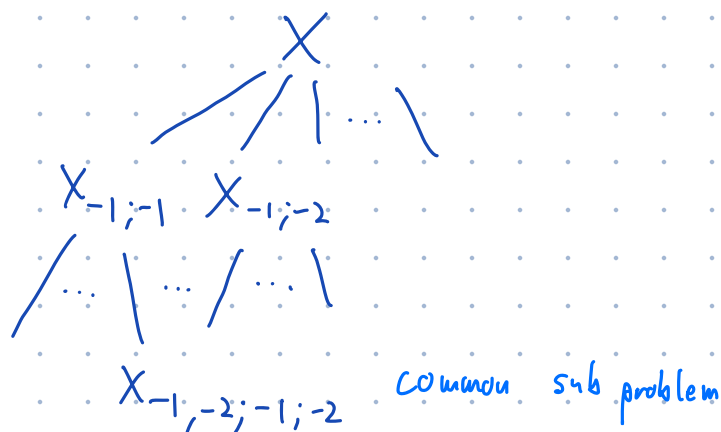
Thm: $\det_n, \text{perm}_n$ have size $O(n \cdot 2^n)$ circuits.

Pf: Use Laplace Expansion:

$$\det_n(X) = \sum_{i=1}^n (-1)^{i-1} x_{1,i} \cdot \det_{n-1}(X_{-1;-i})$$

✗ Inductively, $\det_{n-1}(1, i)$ has $O((n-1) \cdot 2^{n-1})$ circuit ✗

Notice that these have common subproblems:



So for all $S \subseteq [n]$, $S = \{i_1, \dots, i_k\}$, compute in decreasing order of $|S|$

$$\det_{n-k}(X_{-1,-2,\dots,-k;-i_1,-i_2,\dots,-i_k})$$

If you have the smaller subproblems, need $O(n)$ Π 's, single Σ .
Total of 2^n subsets $S \subseteq [n] \Rightarrow$ circuit of size $O(n \cdot 2^n)$.

Have Laplace formula for perm_n (just remove sign) $\Rightarrow$ same result

Thrm: The $n \times n$ determinant, $\det_n$, of $X = (x_{ij})_{1 \leq i, j \leq n}$ has a size $O(n^5)$ circuit

Pf: Use Gaussian elimination to get circuit w/ divisions

$$\begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix} \xrightarrow{O(n) \div's} \begin{bmatrix} 1 & x_{12} & x_{13} \\ \frac{x_{21}}{x_{11}} & x_{22} & x_{23} \\ \frac{x_{31}}{x_{11}} & x_{32} & x_{33} \end{bmatrix} \xrightarrow[\begin{smallmatrix} O(n^2) \times's \\ O(n) \div's \end{smallmatrix}]{O(n^2) \times's} \begin{bmatrix} 1 & x_{12} & x_{13} \\ 0 & x_{22} - \frac{x_{21}x_{12}}{x_{11}} & x_{23} - \frac{x_{21}x_{13}}{x_{11}} \\ 0 & x_{32} - \frac{x_{31}x_{12}}{x_{11}} & x_{33} - \frac{x_{31}x_{13}}{x_{11}} \end{bmatrix}$$

So we can do one column of Gaussian elimination w/ $O(n^2)$ gates
 $\Rightarrow$ size $O(n^3)$ circuit w/ divisions to compute $\det_n$

one band, leave up

How do we eliminate divisions? We use an idea of [Strassen 1973].
Same one who discovered alg to multiply $n \times n$ matrices in $O(n^{\log_2 7})$ mults

The idea* is that $\frac{1}{h} = \frac{1}{1-(1-h)} = \sum_{i=0}^{\infty} (1-h)^i$. $\deg(\det_n) = n \Rightarrow$ we only need up to term $(1-h)^n$

Introduce a new indeterminate z and work in $R_n := \mathbb{F}[X][z] / \langle z^{n+1} \rangle$

Let $Y = I_n - X$. We compute $\det_n(I_n + zY) \in \mathbb{F}[X][z]$
Then at the end, set $z=1$

Elements of the form $1 - z \cdot b(z)$ in R_n are invertible in R_n

$$(1 - z \cdot b(z))^{-1} = (1 - w)^{-1} = (1 + w)(1 + w^2) \cdots (1 + w^{2^k}), \quad k \approx \log(n).$$

So our circuit w/ divisions over $\mathbb{F}$ turns into a circuit w/o divisions over $\mathbb{F}[z] / \langle z^{n+1} \rangle$. We can simulate operations in R_n in $\mathbb{F}[X]$

Simulating Σ in R_n needs $O(n)$ gates in $\mathbb{F}[X]$

Simulating $\Pi, \div$ in R_n needs $O(n^2)$ gates in $\mathbb{F}[X]$

$\Rightarrow \det_n$ has $O(n^5)$ circuit computing it over $\mathbb{F}[X]$

* There are potential definedness issues which I am sweeping under the rug

Why can't we replicate the above for perm_n ? No analogue of Gaussian elimination.

Thm [Bürgisser 2000]: Roughly speaking

if perm_n has $O(\text{poly}(n))$ -sized circuit and the generalized Riemann Hypothesis holds, then $P = NP$.

This leads to the Geometric Complexity Program of Mulmuley and Sohoni in applying tools from representation theory, algebraic combinatorics, and algebraic geometry to try to separate P and NP .

Idea: Study invariants of orbit space of $\det_n, \text{perm}_n$ under action of $GL_n(\mathbb{C})$. These orbit spaces live in $\mathbb{C}^N$, $N = \binom{n^2+n-1}{n}$.

* What I really mean is that the Polynomial hierarchy collapses to the 2nd level.

Sources:

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- Agrawal (2007) - Determinant Versus Permanent
- Agrawal, Kayal, Saxena (2004) - PRIMES is in P
- Bürgisser (2000) - Cook's versus Valiant's hypothesis
- Bürgisser, Clausen, Shokrollahi (1997) - Algebraic Complexity Theory
- Dutta, Jindal, Pandey, Sinhababu (2021) - Arithmetic Circuit Complexity of Division and Truncation
- Horner (1819) - A new method of solving numerical equations of all orders, by continuous approximation
- J. H. van Lint, Wilson (2001) - A Course in Combinatorics
- Ostrowski (1954) - On two problems in abstract algebra connected with Horner's rule
- Shpilka, Yehudayoff (2010) - Arithmetic Circuits: A survey of recent results and open questions
- Shpilka, Wigderson (2001) - Depth-3 Arithmetic Circuits over Fields of Characteristic Zero
- Strassen (1973) - Vermeidung von Divisionen. In: Journal für die reine und angewandte Mathematik 264
- Tutte (1947) - The Factorization of Linear Graphs

Here's a bonus result

Ex [Ben-Or's Trick (1999)]: $e_1, \dots, e_n$ are computed by depth-3 $\Sigma\Pi\Sigma$ circuits of size $O(n^2)$

Pf: Evaluate the following polynomial $F(y) \in \mathbb{C}[x_1, \dots, x_n][y]$ at $\alpha_1, \dots, \alpha_{n+1}$

$$F(y) = \prod_{i=1}^n (y - x_i) = \sum_{i=0}^n (-1)^i y^{n-i} e_i(x_1, \dots, x_n)$$

Let $\beta_i := f(\alpha_i)$

$$\begin{pmatrix} \alpha_1^n & \alpha_1^{n-1} & \dots & \alpha_1 & 1 \\ \alpha_2^n & \alpha_2^{n-1} & \dots & \alpha_2 & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \alpha_n^n & \alpha_n^{n-1} & \dots & \alpha_n & 1 \\ \alpha_{n+1}^n & \alpha_{n+1}^{n-1} & \dots & \alpha_{n+1} & 1 \end{pmatrix} \begin{pmatrix} e_0 \\ -e_1 \\ \vdots \\ e_{n-1} \\ (-1)^n e_n \end{pmatrix} = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \\ \beta_{n+1} \end{pmatrix}$$

LHS is Vandermonde which is invertible

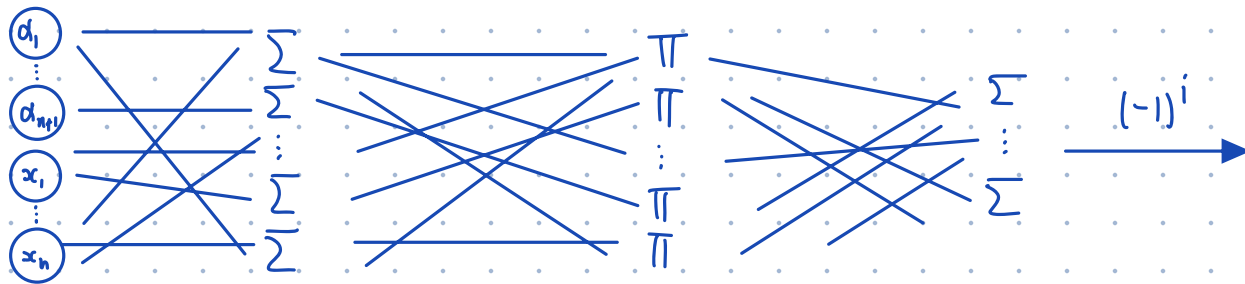
Denote the inverse by $(z_{ij})_{1 \leq i, j \leq n+1}$

With some work, one can find an exact formula for $(z_{i,j})$ in terms of (α_i^{n+1-j})

$$\begin{pmatrix} z_{11} & z_{12} & \dots & z_{1n} & z_{1,n+1} \\ z_{21} & z_{22} & \dots & z_{2n} & z_{2,n+1} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ z_{n1} & z_{n2} & \dots & z_{nn} & z_{n,n+1} \\ z_{n+1,1} & z_{n+1,2} & \dots & z_{n+1,n} & z_{n+1,n+1} \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \\ \beta_{n+1} \end{pmatrix} = \begin{pmatrix} e_0 \\ -e_1 \\ \vdots \\ e_{n-1} \\ (-1)^n e_n \end{pmatrix}$$

one can move this sign inside to get $\Sigma\Pi\Sigma$

$$e_i(x_1, \dots, x_n) = (-1)^i \sum_{j=1}^{n+1} z_{i,j} \prod_{k=1}^n \alpha_j - x_k$$



$O(n^2)$ inputs

compute $\alpha_j - x_k$
 $\Rightarrow O(n^2)$ Σ gates

product of n terms,
do for each j
 $\Rightarrow O(n^2)$ Π gates

sum of $n+1$ terms
 $\Rightarrow O(n)$ Σ gates